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SUPERSEDING
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FEDERAL SPECIFICATION

GRADERS, ROAD, MOTORIZED

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers engine-powered, pneumatic-tired hydraulically-operated motor graders.

1.2 Classification.

1.2.1 Types and sizes. The graders shall be of the following types and sizes as specified (see 6.2):

Type I - 4 wheel, 2 wheel drive, front-wheel steer and articulated frame steer.
Type II - 6 wheel, 4 wheel drive, front wheel steer.
Type III - 6 wheel, 4 wheel drive, front wheel steer and articulated frame steer.
Type IV - 6 wheel, all wheel drive, front wheel steer.
Type V - 6 wheel, all wheel drive, front wheel steer and articulated frame steer.

Sizes - see table I.

TABLE I. Size classification.

Size	Type I	Type II, Type III	Type IV	Type V	Wheelbase, inches, minimum
1	10,000 thru 16,500	10,000 thru 16,500	-	-	154
2	16,501 thru 22,500	16,501 thru 22,500	-	-	198
3		22,501 thru 25,000	-	-	224

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TABLE I. Size classification. (continued)

Size	Type I	Type II, Type III	Type IV	Type V	Wheelbase, inches, minimum
5		27,501 thru 30,500	29,001 thru 30,000	27,501 thru 31,500	233
6		30,501 thru 34,000	30,001 thru 33,000	31,501 thru 35,500	233
7		34,001 up	-	-	240

2. APPLICABLE DOCUMENTS

2.1 Government publications. The issues of the following documents, in effect on date of invitation for bids or solicitation for offers, form a part of this specification to the extent specified herein.

Federal Specifications:

ZZ-T-1083 - Tires, Pneumatic, Low Speed, Off Highway.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions, as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Description from established distribution points in their agencies.)

Military Specifications:

MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment.

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

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MIL-G-762 - Graders, Road, Packaging of.
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty.

Military Standard:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electro-magnetic Interference.

(Copies of specifications, standards, and drawings required by contrctors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Federal Regulations:

29 CFR 1926.600: Equipment.
29 CFR 1926.602: Material Handling Equipment.
29 CFR 1926.1001: Rollover Protective Structure.

(The Code of Federal Regulations [CFR] and the Federal Register [FR] are for sale on a subscription basis by the Superintendent of Documents, U.S. Government printing Office, Washington DC 20402. When indicated reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Department of Transportation (DOT):

DOT Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Society of Automotive Engineers (SAE):

Standards and Recommended Practices.

J53 - Minimum Performance Criteria for Emergency Steering of Wheeled Earthmoving Construction Machines.
J88 - Sounds Measurement - Earthmoving Machinery - Exterior.
J350 - Spark Arrester Test Procedure for Medium Size Engines.

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J551 Limits and Methods of Measurement of Radio Interference Characteristics of Vehicles and Device (20-1000 MHz).
J816 Engine Test Code - Spark Ignition and Diesel.
J833 USA Human Physical Dimensions.
J872 Reserve Tractive Ability Test Code.
J919 Sound Measurement - Earthmoving Machinery - Operator - Singular Type.
J925 Minimum Access Dimensions for Construction and Industrial Machinery.
J1152 Minimum Performance Criteria for Braking Systems for Off-Highway, Rubber-Tired Self-Propelled Construction Machines.
J1166 Sound Measurement - Earthmoving Machinery - Operator - Work Cycle.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

Tire and Rim Association, Inc. (TRA):

Tire and Rim Association Yearbook.

(Application for copies should be address to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

American Society For Testing and Materials (ASTM):

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Description. The graders shall be diesel-engine-driven, pneumatic-tired and operable by one operator for general construction work, such as course and fine grading, low and high bank cutting, scarifying and V-ditching. The graders shall be equipped with all necessary operating accessories customarily furnished whether stipulated herein or not, together with such modifications and attachments as may be necessary to enable the unit to function reliably and efficiently in sustained operation. All types of the grader shall have leaning front wheels. All movement of the moldboard, except moldboard tilt on size 1 graders, scarifier and leaning wheel mechanisms shall be controlled through hydraulic valves.

3.2 Material. Material shall be as specified herein (see 6.6). Material not specified shall be the manufacturer's current offering to the public differing herefrom only in respects necessary to meet special requirements. Components, particularly of the electrical, fuel and exhaust systems shall resist harmful effects of dust, moisture and corrosion.

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3.2.1 Material deterioration and control. The item(s) shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the item may be exposed.

3.2.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.2.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative

3.3 Grader characteristics.

3.3.1 Weight. The operating weight of the grader shall be as specified in table I. Weight shall be determined with the grader in a fully serviced condition including fuel, oil, water, rollover protective structure (ROPS) cab, and with standard accessories as specified herein. Weight shall not include a scarifier or ripper, hydroinflation or other tire ballast, bolt on wheel weights, or other bolt on "dead" weight, or an operator. The grader shall have not less than 65 percent nor more than 75 percent of the operating weight on the rear tandem wheel except type I.

3.3.2 Dimensions. The minimum wheelbase shall be as specified in table I.

3.3.3 Type I. The type I graders shall be of the 4 wheel, 2-wheel drive, front-wheel steer, articulated frame type. The rear drive shall be rigid and shall have an automatic locking differential.

3.3.4 Type II. The type II graders shall be of the 6-wheel, 4-wheel drive, front-wheel steer, rigid frame type. The final drive shall be on the four rear wheels, arranged in tandem, two on each side. The tandem frames shall be mounted on and pivot about the driving axle so that equal weight is carrier on all four wheels when operating over rough terrain to minimize the affect on the evenness of the blade cut.

3.3.5 Type III. Type III graders shall be of the 6-wheel, 4-wheel drive, front-wheel steer, articulated frame type. The final drive shall be on the four rear wheels, arranged in tandem, two on each side. The tandem frames shall be mounted on, and pivot about, the driving axle so that equal weight is carried on all four wheels operating over rough terrain to minimize effect on the evenness of the blade cut. The grader shall have a limited slip differential or a manual lock-unlock differential or an automatic locking differential.

3.3.6 Type IV. Type IV graders shall be of the 6-whee1, all wheel drive, front-wheel steer, rigid frame type. The tandem drive frames shall be mounted

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on and pivot about the driving axle so that equal weight is carried on all four wheels when operating over rough terrain to minimize affect on the evenness of the blade cut.

3.3.7 Type V. Type V graders shall be of the 6-wheel, all wheel drive, front-wheel steer, articulated frame type. The tandem drive frames shall be mounted on and pivot about the driving axle so that equal weight is carried on all four wheels when operating over rough terrain to minimize effect on the evenness of the blade cut. The grader shall be equipped with an automatic or manual locking differential.

3.4 Frame. The frame shall be of structural steel. The frame shall be designed to withstand maximum stresses under normal operating conditions, and in addition, provide adequate support for attaching any device approved by the grader manufacturer for use in combination with the grader. Articulated frames shall be joined by a rugged hinge assembly that allows free movement to either side but which holds the frame rigid against twisting forces. A mechanical lockout to prevent steering action shall be provided on articulated frames.

3.5 Safety. All available standard options and accessories shall be furnished to provide protection to maintenance and operating personnel from rotating or reciprocating parts conforming to 29 CFR 1926.1001 and from parts that are subject to high operational temperatures and so located as to be a hazard. Protective devices shall not impair operating functions. The starting system shall be such that the transmission must be placed in neutral or the master clutch disengage before the engine can be started or the transmission must be placed in neutral before the grader can be set in motion. Graders with articulated frames shall have a decal located in the area of the articulation points warning of personnel clearance during frame articulation and placement of the safety pin or bar in the transport mode. Handles and nonskid steps shall be of a size, shape and location to permit safe mounting and dismounting. The graders shall be furnished with seatbelts, horn and reverse signal alarm conforming to 29 CFR 1926.602. Unless otherwise specified herein, the graders shall conform to all Federal Laws and Regulations governing noise, air pollution and other health and safety requirements which are in effect for this type of equipment at the time of contract award.

3.5.1 Noise. The grader shall not exceed 88dB(A) when tested for exterior sound level as specified in 4.2.4. The grader should not exceed 85dB(A) when tested for operator sound exposure as specified in 4.2.4. If operator sound exposure exceeds 85dB(A) for any part of specified test, a noise hazard decal stating at least the following shall be permanently mounted in the operator compartment: "Caution. Hearing Protection Required".

3.6 Ease of maintenance. The design of the graders shall permit ready accessibility for servicing, replacement and adjustment of component parts and accessories with minimum disturbances to other elements for the small person to the large person arctic clothed as defined by SAE J833 and J925.

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3.7 Engine. The engine shall be the heavy-duty diesel furnished commercially including all systems, components, accessories and auxiliaries. The engine shall be either two- or four-stroke cycle, dependent upon normal practice of the grader manufacturer.

3.7.1 Cooling system. The manufacturer's current standard production engine cooling system shall be provided. A radiator and fan tip guard shall be provided. The cooling system shall be filled with a clean solution of 50 percent water and 50 percent ethylene glycol type antifreeze conforming to MIL-A-46153.

3.7.2 Engine starting requirements. The engine shall start within 5 minutes and the grader controls shall be operable within 15 minutes after engine start in any ambient temperature in the range of -25° F to +110° F. The engine shall have glow plugs or a premeasured shot, ether type cold start aid controlled from the operator's seat. If either type start aid is used, start aid shall not disperse ether when engine or ambient temperature does not warrant starting aid use.

3.7.3 Engine shields. The engine shall be protected with metal or fiberglass top and lower hood side panels. When the grader frame members or other components provide protection for the engine, lower side panels will not be required. When specified (see 6.2), the graders shall be equipped with removable side panels with quick-disconnect latches or vertical hinged engine compartment doors.

3.7.4 Lubricating system. The manufacturer's current standard production engine lubricating system shall be provided.

3.7.5 Engine accessories. The engine shall be equipped with the following:

a. Two stage dry type air cleaner with service indicator and automatic dust ejector.
b. Oil Filter, full-flow replaceable-element type.
c. Fuel filter, full-flow replaceable-element type.
d. Governor.
e. Muffler.
f. Exhaust stack with rain cap or curved bevel-cut extension.
g. When specified (see 6.2), nonturbocharged engines shall be equipped with a spark arrester having an arresting efficiency of 80 percent as determined by SAE J350.

3.7.6 Fuel tank. The fuel tank shall be of a type regularly furnished as standard equipment with a capacity of 8 hours of continuous operation and shall be equipped with means for draining water and sediment from the bottom of the tank.

3.8 Blade pull. When the graders are tested as specified in 4.2.2, the curve through the maximum power points for successive gears shall be on or above the

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applicable curve shown on figure 1. Data shall be taken to establish the points of maximum blade horsepower in each gear (which are the inflection points in the curves of each of the gears in most transmissions). An overall curve drawn through the average values of these maximum blade horsepower points shall be coincident with, or to the right of, the requirement (see figure 1) for the range of ground speeds for 2.0 to 10.0 mph. In recording blade pull data, only those points (blade pull vs ground speed) with slip values of 15 percent or less shall be valid.

3.9 Transmission.

3.9.1 Powershift transmission with or without torque converter. Unless otherwise specified (see 6.2), the grader shall be equipped with a powershift type transmission and shall be furnished with one of the following shifting arrangements, dependent upon normal practice of the grader manufacturer, described in a, b, or c below:

a. Shifting under full engine power through successive forward and reverse gear ratios.
b. Shifting under full engine power through successive forward and reverse gear ratios within ranges.
c. Shifting front forward to reverse or vice versa by a rocker pedal in lieu of a clutch pedal without hand shifting.

One of the shifting arrangements described in a, b, and c above may be selected. Graders equipped with torque converter drives shall have gear reductions suitable to keep the torque converter working efficiently throughout the entire range of grader speeds. The number of forward and reverse speeds, the maximum forward speed ranges in first and second gears in the unloaded condition, and the minimum travel speed in high gear for each size of grader at governed engine speed, shall be as shown in table II.

3.9.2 Friction clutch constant mesh type transmission. When specified (see 6.2), the transmission shall be of the constant mesh type for successive shifting of forward and reverse gears and shall be equipped with a power input shaft clutch brake to retard transmission gear rotation when shifting from one gear to another. Transmission of the sliding gear type will be acceptable for size 1 graders. The number of forward and reverse speeds and the maximum forward speed ranges in first, second, and third gears in the unloaded condition and the minimum travel speed in high gear for each size of grader at governed engine speed shall be as shown in table III.

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TABLE II. Speed requirements for powershift equipped graders - speeds (mph).

Size (all types)	Minimum number of speeds FWD	REV	Reverse speed ranges lowest speed	highest speed	Forward speed ranges 1st gear	2nd gear	high gear (minimum)
1	3	3	0 to 4.1	7.0 and up	0 to 4.0	2.7 to 8.0	18
2	3	3	0 to 4.0	7.0 and up	0 to 4.0	2.7 to 8.0	18
3	4	4	0 to 4.0	7.0 and up	0 to 4.0	2.7 to 8.0	20
4	4	4	0 to 4.0	7.0 and up	0 to 4.0	2.7 to 8.0	20
5	4	3	0 to 4.0	7.0 and up	0 to 4.0	2.7 to 8.0	20
6	4	3	0 to 4.0	7.0 and up	0 to 4.0	2.7 to 8.0	20
7	4	4	0 to 4.0	7.0 and up	0 to 4.0	2.7 to 8.0	20

TABLE III. Speed requirements for friction clutch equipped graders - speeds (mph).

Size (all types)	Minimum number of speeds FWD	REV	Reverse speed ranges lowest speed	highest speed	Forward speed ranges 1st gear	2nd gear	3rd gear	high gear (minimum)
1	4	4	0 to 4.0	8.0 and up	1.6 to 3.5	2.7 to 7.0	8.2 to 13	18
2	4	4	0 to 4.0	8.0 and up	1.6 to 3.5	2.7 to 7.0	8.2 to 13	18
3	8	4	0 to 4.0	8.0 and up	1.9 to 3.0	2.8 to 4.8	4.0 to 7.8	20
4	8	4	0 to 4.0	8.0 and up	1.9 to 3.0	2.8 to 4.8	4.0 to 7.8	20
5	8	4	0 to 4.0	8.0 and up	1.9 to 3.0	2.8 to 4.8	4.0 to 7.8	20
6	8	4	0 to 4.0	8.0 and up	1.9 to 3.2	2.8 to 4.8	4.0 to 7.8	20
7	8	4	0 to 4.0	8.0 and up	1.9 to 3.2	2.8 to 4.8	4.0 to 7.8	20

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3.10 Master clutch. Graders, other than those equipped with a powershift transmission, shall be equipped with a heavy-duty oil type master clutch (except size 1 graders). The clutch shall be adjustable for wear and shall have the capacity to transmit full power of the engine under maximum load conditions without slippage or evidence of overload.

3.11 Final drive. All shafts, gears, sprockets, chains, and bearings shall have sufficient strength and capacity to safely transmit full power of the engine to the driving wheels. Drive gears shall run in an oil bath and all driving mechanisms shall be fully enclosed and protected against lubricant leakage and entrance of dust, mud, and water.

3.12 Limiting device. All graders, except type I, shall have a device permanently attached to the grader to limit or prevent tandem oscillation. In lieu of the limiting device, lifting attachments may be located at the center of balance of the tandem cases to prevent tandem case oscillation or damage to the grader during loading and shipment.

3.13 Front axle. The front axle shall be pivot mounted, of rugged construction, and fitted with leaning-type wheels and operating mechanism. Positive means shall be provided to prevent the tires from rubbing when the wheels are in the leaning position. The leaning wheel mechanism shall lean the wheels not less than 15 degrees right and left.

3.14 Steering. The graders shall have hydraulically boosted mechanical or full hydraulic power steering. The grader shall be equipped with an emergency steering system conforming to SAE J53 that will give the operator steering control in the event of engine failure.

3.15 Controls. All movement of the moldboard (except moldboard tilt size 1 grader), scarifier and leaning wheel mechanism shall be controlled through hydraulic valves. All control levers shall be located in and operated from the operator’s compartment. The controls shall permit simultaneous-operation of not less than three functional components without appreciable loss of component speed or power.

3.16 Hydraulic system. The hydraulic system shall be the contractor’s standard commercial system and shall provide sufficient force and means for operating the blade, scarifier, and related components.

3.17 Electrical system. Unless otherwise specified (see 6.2), the manufacturer's standard commercial electrical system shall be provided. The battery terminal cable shall be provided with nonconductive boots to prevent accidental shorting. The battery circuit shall be provided with a reverse current device if required to protect the voltage regulator from damage when the batteries are connected in reverse polarity.

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3.18 Brake system. A service brake system effective on all four wheels on the tandem axles shall be provided. A parking brake and an emergency stopping system shall be provided. The braking system shall comply with SAE J1152.

3.19 Operator's compartment.

3.19.1 Cab. Unless otherwise specified (see 6.2), a fully enclosed ROPS cab shall be furnished. The cab shall comply with 29 CFR 1926.1001 (ROPS) and conform to 29 CFR 1926.600 (glass). The cab shall be the manufacturer's current standard production type, enclosing the operator's seat and all controls. The cab shall not obstruct the view of operation while the operator is seated. The cab shall be equipped with two doors, one on each side of the cab, and each door shall be provided with a lock or latch to secure the door in an open position. A personnel heater, defroster fan, front and rear windshield wipers and a rear view mirror (mounted outside and on the left hand side of the cab) shall be furnished.

3.19.2 Seat. Unless otherwise specified (see 6.2), the manufacturers standard commercial operator's seat shall be provided. The seat shall be located to provide the operator with convenient access to operating controls and to provide good visibility of the work zone, when seated with seatbelt fastened and properly adjusted.

3.19.3 Instruments and gages. The manufacturer’s standard instruments and gages shall be provided. In addition to the standard instruments and gages furnished, any or all of the following instruments and gages shall be furnished as specified (see 6.2):

a. Engine lubricating oil pressure.
b. Engine cooling liquid temperature.
c. Torque converter temperature (when applicable).
d. Ammeter, voltmeter or battery charging indicator.
e. Hourmeter.
f. Odometer.
g. Speedometer or tachometer.
h. Fuel level indicator.

3.20 Circle and moldboard assembly. The circle and moldboard assembly shall permit the grader to perform all operations listed in table IV. The moldboard sideshift shall be hydraulic power operated, both to the right and to the left of the circle. The assembly shall be provided with shims or other means of adjustment for wear.

3.20.1 Moldboard. Unless otherwise specified (see 6.2), moldboards shall be in accordance with table IV, and shall be equipped with standard punched replaceable cutting edges having a high resistance to wear and abrasion. Cutting edges shall be of a commercial type.

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TABLE IV. Moldboard minimum dimensions.

Size	Length (feet)	Thickness (inches)	Width with cutting edge[1/] (inches)	Cut below ground level[2/] (inches)	Lift above ground level[2/] (inches)	Reach outside rear tires[2/] right & left (inches)	Bank cutting angle right & left (degrees)	Rotation (degrees)
1	10	1/2	14	4	10	36-36	-	120
2	12	3/4	22	12	13	65-60	90	360
3	12	3/4	24	16	16	68-68	90	360
4	12	3/4	24	17	16	68-68	90	360
5	12	3/4	24	17	16	72-68	90	360
6	12	7/8	24	17	16	72-68	90	360
7	14	1	27	17	16	82-81	90	360

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3.20.2 Actuating mechanism interference. No movement or combination of movements or adjustments of the moldboard assembly shall cause any part of the actuating mechanism to contact any structural part of the grader in such a manner as to damage the structure or mechanism when operated in accordance with approved operating and maintenance manuals. Electric limit switches to provide this protection shall not be permitted.

3.21 Wheels and tires. Commercial standard remountable wheels or rims shall be provided. Tires shall be of the tubeless type and shall have a tread pattern suitable for grader operation. Tires shall be in accordance with ZZ-T-1083. Unless otherwise specified (see 6.2), tires shall be not less than the size and ply rating shown in table V. All valve parts shall conform to the Tire and Rim Association Yearbook.

TABLE V. Tires.

Grader size	Minimum tire size	Ply rating
1	38.5 x 15	6
2	13.00 x 24	8
3	13.00 x 24	8
4	13.00 x 24	10
5	13.00 x 24	12
6	14.00 x 24	10
7	16.00 x 24	12

3.22 Tow hitch. Graders shall be equipped with the manufacturer's standard rear hitch. The height from the ground to the centerline of the hitch jaw opening shall be between 20 and 38 inches.

3.23 Tools and toolbox.

3.23.1 Tools. When specified (see 6.2), the following listed tools and equipment shall be furnished with each grader:

One - hydraulic jack, 7-ton capacity.
Two - chisels, cold, 3/4 by 8 inches long.
One - hammer, ball-peen, 3-pound size.
One - pliers, combination slip joint, 8-inch size.
One - pliers, diagonal cutting, 7-inch size.
One - punch, tapered, 10 inches long.
One - screwdriver, 6 by 3/16 inch.
One - screwdriver, 10 by 1/4 inch.
One - screwdriver, 12 by 5/16 inch.
One - wrench, front wheel spindle nut.

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One - wrench, wheel stud nut (when applicable).
One - wrench, adjustable crescent, 6-inch.
One - wrench, adjustable crescent, 12-inch.
One - oiler, hand, l-pint size.
One - grease gun, high-pressure, level type of 16-ounce capacity.

3.23.2 Toolbox. The grader shall be equipped with the manufacturer's standard or optional toolbox, as applicable. When the above tools are furnished the toolbox shall be of a size and shape to hold all tools provided.

3.24 Optional accessories and attachments. When specified (see 6.2), graders shall be furnished with the following or other accessories and attachments as specified (see 6.2):

a. The following lights (meeting DOT Motor Carrier Safety Regulations, as applicable) with associated circuits and switches shall be provided:

   1. Two headlights.
   2. Two working lights (mounting location shall be furnished by buyer).
   3. Combination stop and taillights.
   4. Turn signals with emergency flasher.
   5. Instrument panel lighting.
   6. Rear floodlight.
   7. Dome light (cab).
   8. Flashing amber beacon rotating, mounted togive 360 degree visibility.

b. Scarifier. The grader shall be furnished with a V-type scarifier equipped with not less than 9 removable shanks (7 for size 1 graders) with replaceable points.
c. Ripper. When a scarifier is required, a rear mounted ripper shall be furnished in lieu thereof when that is the contractor's standard practice. The ripper shall be equipped with not less than 3 replaceable shanks with replaceable points.
d. Bulldozer. The bulldozer shall be equipped with standard punched replaceable cutting edges having a high resistance to wear and abrasion. Cutting edges shall be of a commercial type. The blade shall be equipped with an all-welded steel push frame for mounting on the grader and may be controlled by either the scarifier mechanism or an independent mechanism.
e. Snowplow. The snowplow shall be of the V-blade type and shall be equipped with replaceable cutting edges. The snowplow shall be provided with replaceable adjustable shoes with renewable wear plates.
f. Snow wing. The snow wing shall be not less than 12 feet long and shall be provided with adjustable arms for setting the horizontal angle of the wing to at least the angle the V-plow makes with the horizontal axis of the grader. Vertical adjustment shall be accomplished hydraulically or by other suitable methods with controls accessible to the operator in his normal operating position. Cutting edges conforming to the contractor’s standard commercial product shall be provided.

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g. Blade extensions, right and left.
h. Front tow hook(s) or eye(s).
i. Radiator shutters.
j. Hydraulic tilt moldboard (size 1 grader).
k. Anti-vandalism protection. The grader shall be equipped with locks to secure the following:

1. Fuel tank filler cap.
2. Hydraulic oil tank filler cap.
3. Engine oil filler cap and distick.
4. Transmission oil filler cap and dipstick.
5. Radiator filler cap.
6. Toolbox.
7. Battery box.
8. Cab doors.

When any of the above items are secured by lockable cab doors and lockable engine side panels, individual locks are not required.

l. Automatic blade control.
m. Fire extinguisher. Minimum 10 ABC rating.
n. Lifting attachments. Permanently installed lifting attachment slings.
o. Tire inflation kit or pump. The tire inflation kit or pump shall be equipped with flexible high pressure air hose of sufficient length to reach all wheels. The hose shall be equipped with a valve stem chuck and tire pressure gage.
p. Storage box. The storage box shall be of sufficient size to store the scarifier shanks with the points installed.
q. Additional accessories and attachments as specified (see 6.2).

3.25 Special requirements.

3.25.1 Electromagnetic interference characteristics. When specified (see 6.2) the grader shall conform to the electromagnetic interference control characteristics of SAE J552 or electromagnetic interference control requirements of MIL-STD-461 for class IIIC equipment.

3.25.2 Fungus resistance. When specified (see 6.2), the electrical components of the grader shall be coated with varnish conforming to MIL-V-173, except that: (a) Components and elements inherently inert to fungi or hermetically sealed enclosures need not be coated, (b) Current-carrying surfaces, such as relay contact points, shall not be coated.

3.26 Painting. The portions of the grader normally painted shall be cleaned, treated, primed, and painted in accordance with the contractors standard commercial practice. Unless otherwise specified (see 6.2), the finish color shall be either highway yellow or the contractor’s commercial yellow.

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3.27 Identification plate. When specified (see 6.2), an identification plate conforming to MIL-P-514, shall be permanently affixed to the grader in a conspicuous place. The contractor shall stamp all necessary data in the blank spaces provided for that purpose.

3.28 Manuals. Unless otherwise specified (see 6.2), the contractor shall furnish with each grader one copy each of the operators, maintenance, repair parts and overhaul manuals to cover the entire grader.

3.29 Precautionary notes, warnings and use instructions. In addition to furnishing written operating instructions as specified in 3.28, the contractor shall provide instruction or special caution plates or decals containing instructions or suitable warnings wherever deemed advisable by Government or the contractor in accordance with standard commercial practice.

3.30 Data questionaire. When specified (see 6.2), a completed questionnaire, appendix A of this specification, shall be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection, tests and operation.

4.2.1 Inspection. One grader from each contract shall be examined to determine proper functionality and compliance with the requirements marked “X” in columns 1 and 3 of table VI. Each grader shall be examined to determine proper functionality and compliance with the requirements marked “X” in column 2 of table VI. Nonconformance to one or more requirements shall be cause for rejection.

4.2.2 Tests. One grader from each contract shall be tested to determine compliance with 3.8. Compliance with 3.8 shall be determined from pull tests in accordance with SAE J872, except that the load inducing cable shall be carried horizontally 6 to 9 inches above the groundline. Maximum wheel slippage allowed is 15 percent. Nonconformance to 3.8 shall constitute failure of this test.

4.2.3 Operation. Start and operate each grader for not less than 1 hour. Manipulate all controls and adjustable devices. Inability of the grader to start and operate for 1 hour or failure of any control or adjustable device shall be cause for rejection.

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4.2.4 Noise testing. Exterior sound level shall be measured in accordance with SAE J88. Nonconformance to 3.5.1 exterior sound requirement shall constitute failure of this test. Operator sound level exposure shall be tested in accordance with SAE J1166 and J9190

4.3 Certification.

a. Manufacturer's signature upon written notice shall signify acceptable certification of compliance with designated characteristics in column 3 of table VI.
b. In lieu of pull tests required in 4.2.2, certified test reports or certified data may be provided to prove compliance with 3.8.
c. In lieu of noise testing required in 4.2.4, certified test reports or certified data may be provided to prove compliance with 3.5.1.

TABLE VI. Inspection schedule.

1	2	3	Characteristics	Requirement paragraph(s)
-	X	-	101. Grader type not as specified.	1.2.1, 3.3.3 thru 3.3.7
-	-	X	102. Grader size and operating weight not as specified.	1.2.1, 3.3.1
X	-	-	103. Material not as specified.	3.2
X	-	-	104. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.2.1
X	-	-	105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.2.2
X	-	-	106. Contractor does not have documentation available for identification of material, material finishes or treatment.	3.2.3
X	-	-	107. Dimensions not as specified.	3.3.2, 3.20, 3.20.1
X	-	-	108. Frame not as specified.	3.4
-	-	X	109. Safety components not as specified.	3.5, 3.5.1
-	-	X	110. Ease of maintenance not as specified.	3.6
X	-	-	111. Engine not as specified.	3.7
X	-	-	112. Engine cooling system not as specified.	3.7.1
-	-	X	113. Engine starting and fluid priming system not as specified.	3.7.2
-	X	X	114. Engine shields not as specified.	3.7.3
-	X	X	115. Engine lubricating system not as specified.	3.7.4

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TABLE VI. Inspection schedule. (Continued)

1	2	3	Characterisitcs	Requirement paragraph(s)
X	-	-	116. Engine accessories not as specified.	3.7.5
-	-	X	117. Fuel tank not as specified.	3.7.6
-	-	X	118. Powershift transmission not as specified.	3.9.1
-	-	X	119. Friction clutch constant mesh type transmission not as specified.	3.9.2
-	-	X	120. Master clutch not as specified.	3.10
-	-	X	121. Final drive not as specified	3.11
X	-	-	122. Limiting device not as specified.	3.12
X	-	-	123. Front axle and wheel lean not as specified.	3.13
-	X	-	124. Steering not as specified.	3.14
-	X	-	125. Controls not as specified.	3.15
-	-	X	126. Hydraulic system not as specified.	3.16
-	-	X	127. Electrical system not as specified.	3.17
-	-	X	128. Brake system not as specified.	3.18
-	-	X	129. Cab not as specified.	3.19.1
X	-	-	130. Operator's seat not as specified.	3.19.2
X	-	-	131. Instruments and gages not as specified.	3.19.3
X	-	-	132. Circle and moldboard assembly not as specified.	3.20
-	-	X	133. Cutting edges not as specified.	3.20.1
-	-	X	134. Actuating mechanism not as specified.	3.20.2
X	-	-	135. Tire size and ply rating not as specified.	3.21
X	-	-	136. Wheels not compatible with tires furnished.	3.21
X	-	-	137. Tow hitch not as specified.	3.22
-	X	-	138. Tools and toolbox not as specified.	3.23.1, 3.23.2
-	X	-	139. All optional accessories and attachments ordered not as specified.	3.24
-	-	X	140. Radio interference suppression not as specified.	3.25.1
-	-	X	141. Fungus resistance not as specified.	3.25.2
-	X	-	142. Paint not as specified.	3.26
-	X	-	143. Identification plates not as specified.	3.27
-	X	-	144. Manuals not as specified.	3.28
-	X	-	145. Precautionary notes, warning and instruction plates or decals not furnished.	3.29

Column 1: Inspection conducted on one grader for each contract.
Column 2: Inspection conducted on all graders for each contract.
Column 3: Inspection conducted on one grader for each contract or manufacturer's written certification of compliance.

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4.4 Inspection of preparation for delivery. The preservation, packing, and marking shall be examined in accordance with the quality assurance provisions of MIL-G-762.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or B, or commercial as specified (see 6.2).

5.1.1 Level A. Each grader and its accessories shall be preserved in accordante with the level A requirements of MIL-G-762.

5.1.2 Level B. Each grader with its accessories shall be preserved in accordance with the level B requirements of MIL-G-762.

5.1.3 Commercial. Each grader and its accessories shall be preserved in accordance with ASTM D 3951. In addition, the engine crankcase shall be filled to the operating level with the prescribed lubricating oil; and the cooling system shall be filled with a solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153.

5.2 Packing. Packing shall be level A or B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete grader shall be packed in accordance with the level A packing requirements of MIL-G-762. Tools shall be stowed in the tool box provided; the tool box shall be secured closed to prevent undetected pilferage.

5.2.2 Level B. Each complete grader shall be packed in accordance with the level B packing requirements of MIL-G-762. Tools shall be packed as specified in 5.2.1.

5.2.3 Commercial. Each complete grader shall be packed for mobile shipment in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Levels A and B. Marking of levels A and B shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking of commercial preservation and packing shall comply with ASTM D 3951. In addition, weight and cube data shall be marked on all containers.

6. NOTES

6.1 Intended use. Where factors such as weather, tractive conditions, and other unusual work conditions prevail, the agency issuing the bids should describe such conditions of intended use.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type and size required (see 1.2.1).
c. When removable engine side panels with quick-disconnect latches or vertical hinged engine compartment doors are required (see 3.7.3).
d. When a spark arrester is required (see 3.7.5 g).
e. When a powershift type transmission is not required (see 3.9.1).
f. When a particular shifting arrangement is desired (see 3.9.1 a, b, or c.)
g. When a constant mesh type transmission is required (see 3.9.2).
h. When an electrical system other than specified is required (see 3.17).
i. When a ROPS canopy conforming to 29 CFR 1926.1001 is required (see 3.19.1).
j. When an operator's seat other than specified is required (see 3.19.2).
k. Instruments and gages required in addition to standard equipment (see 3.19.3).
l. When moldboards other than specified are required (see 3.20.1).
m. When tires other than specified are required (see 3.21).
n. When tools are required (see 3.23.1).
o. When optional accessories and attachments are required. The purchaser should specify which accessories and attachments are required (see 3.24 a through p), and all additional accessories and attachments required (see 3.24 q).
p. When radio interference suppression is required (see 3.25.1).
q. When treatment for fungus resistance is required (see 3.25.2).
r. When finish color other than specified is required (see 3.26).
s. When identification plates are required (see 3.27).
t. When manuals are not required or when more than one set of manuals is required (see 3.28).
u. When a completed data questionnaire is required (see 3.30 and 6.8).
v. Degree of preservation and packing required (see 5.1 and 5.2).
w. When contractor's service is required (see 6.5).

6.3 Warranty. The successful bidder shall be required to guarantee the motor graders and parts thereof against defective material and workmanship for a period of 12 months or 2,000 hours, whichever occurs first, from date of acceptance of the equipment. On equipment used within the 50 states of the United States and the District of Columbia, the guarantee shall include the furnishing without cost to the Government, F.O.B. manufacturer's nearest dealer or branch, or to the original destination if desired by the Government, of new parts and assemblies to replace any that prove to be defective within the warranty period. In addition, when the Government elects to have the work performed by the contractor, the cost of the labor involved in the replacement of the defective parts or assemblies, shall be borne by the contractor. The contractor shall bear all costs of transporting the grader to his branch headquarters or the travel cost necessary to send qualified service people to the grader location to make

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repairs under warranty. On equipment used outside the 50 states of the United States and the District of Columbia, the guarantee shall include the furnishing of new parts or assemblies to replace any returned to the contractor by the Government which prove to be defective. The replacement parts or assemblies shall be delivered by the contractor to the port of embarkation in the United States designated by the Government. The contractor shall not be required to bear the cost of labor involved in correcting defects in equipment used outside the 50 states of the United States and the District of Columbia. The contractor shall further certify that the motor graders conform to all federal laws and regulations governing noise, pollution, and other health and safety requirements which are required for the manufacturer for this type of equipment at the time of contract award (see 4.3).

6.4 Repair parts and service availability and bid qualification. If available service information is desired, invitations to bid should require data on repair parts, and services, as included in 6.4.1.

6.4.1 Repair parts and service. As the continuous operation of the grader contemplated by this specification is of utmost importances it is necessary that the successful bidder be in a position to render prompt service and to furnish replacement parts. Accordingly, bidders shall indicate the extent of their ability to render prompt service by furnishing the location of their nearest dealer and branch office to the requisitioning activity where stocks of repair parts are maintained and can be secured within a reasonable time after ordering by part number from the parts book.

6.5 Manufacturer's field maintenance. When specified (see 6.2), upon receipt of the equipment at the destination, the contractor shall furnish to the Government at no extra cost the services of a fully qualified, trained mechanic or serviceman to inspect arid insure the proper assembly and operation of the equipment. It is required that the mechanic or serviceman completely adjust the grader, install all components, lubricate the equipment with the grade of lubricants recommended by the manufacturer, charge the battery and make the unit ready for continuous operation. In addition, the Government’s operator shall be instructed in the proper operation and maintenance of the machine. All of the materials required for the foregoing operations shall be furnished by the contractor. The contractor shall also furnish a minimum of one free follow-up service call to be made by a competent, fully qualified, trained mechanic or serviceman. This service call shall be made at such time, either within 45 days after complete delivery of the equipment or after 30 days’ usage as may be designated after such delivery by the contracting officer or his duly authorized representative.

6.6 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the grader may

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be newly fabricated from recovered materials to the maximum extent practicable, provided the grader produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the grader.

6.7 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

6.8 Data questionaire. When a completed data questionnaire (see appendix) is required for bid evaluation, the contracting officer should include appropriate clauses in the solicitation-

MILITARY INTEREST:

Custodians

Army - ME
Navy - YD

Review activities

Army - AT
Navy - MC
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITIES:

DCG
NHT
FSS
BLM
AFS

PREPARING ACTIVITY:

Army - ME

Project 3805-0119

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein.

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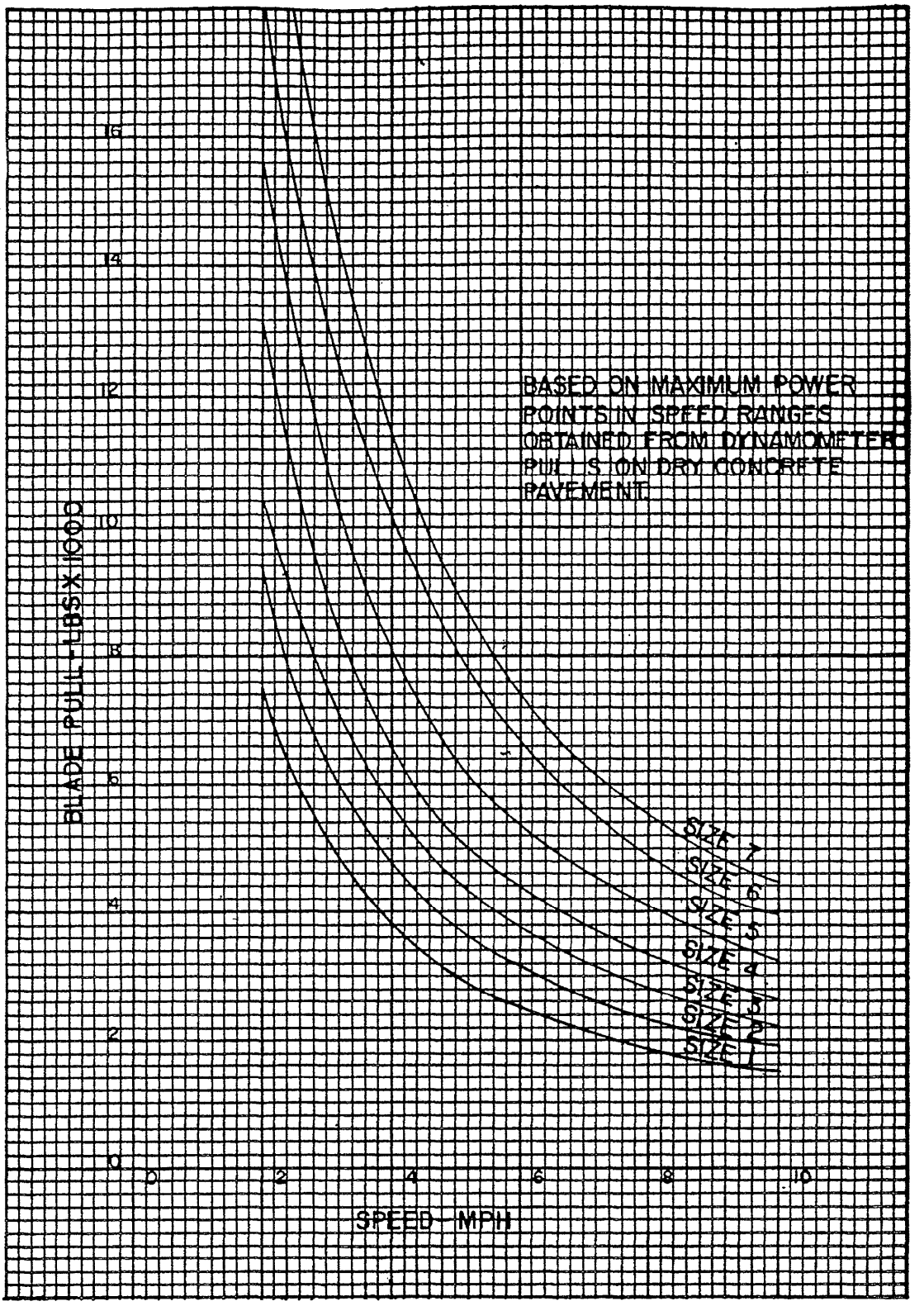


FIGURE 1. Minimum tractive effort requirements (all types).

X-1937A

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APPENDIX A

MOTOR GRADER AND PRODUCT DESCRIPTION

AND

PHYSICAL AND FUNCTIONAL DIMENSIONS

Part A. Description

Manufacturer_____

Manufacturer's Model_____

Type and Size_____

I. Engine

a. Make_____ b. Model_____

c. Series_____ d. Type_____

e. Cycle_____ f. No. of cylinders_____

g. Bore_____ h. Strokes_____

i. Injector size_____ j. Compression ratio_____

k. Total displacement (cu in)_____

l. Rotation (from flywheel end)_____

m. Air induction system_____

n. Net flywheel horsepower_____ at _____ governed rmp.

o. Turbo charged. Yes_____ No_____

Determined by manufacturer with all accessories necessary for engine to perform its intended function in accordance with SAE J816.

II. Engine accessories.

1. Air cleaner.

a. Make_____ b. Model_____ c. Type_____

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2. Oil filter.

a. Type_____ b. Make_____ c. Element No._____

3. Fuel filter.

Primary:

a. Type_____ b. Make_____

Secondary:

a. Type_____ b. Make_____

4. Engine cooling temperature gage.

a. Type_____ b. Make_____ c. Range_____° F

to _____° F d. Location_____

5. Hourmeter - "A" unit.

a. Type_____ b. Make_____

c. Location_____ d. Model_____

Hourmeter - "B" actuator.

a. Make_____ b. Model_____

6. Governor.

A. Unit.

a. Type_____ b. Make_____ c. Model_____

d. Setting for_____engine rpm.

B. Tail shaft.

a. Type_____ b. Make_____ c. Model_____

d. Setting for_____engine rpm.

7. Muffler.

a. Type_____ b. Make_____

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8. Rain cap_____

9. Lubricating oil pressure gage.

a. Type_____ b. Make and model_____

c. Range_____ to _____psi

10. Hand throttle.

a. Type_____ b. Location_____

III. Cooling system.

1. General.

a. Type_____ b. Overall capacity_____gal

c. Pressurization_____psi

2. Radiator.

a. Type_____ b. Capacity_____gal

c. Make_____ d. Serial No._____

e. Protective guard_____

3. Fan.

a. Type: Mechanical_____ Electrical_____

b. Diameter_____inches c. No. blades_____

d. Fan: Push_____ Pull_____

e. Control: Temperature_____ Manual_____

4. Water pump.

a. Type_____ b. Make and Model_____

c. Drive_____ d. Rating_____gpm

5. Thermostat.

a. Type_____ b. Make and model_____

c. Temperature rating: Full open_____° F; closed_____° F

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IV. A. Transmission.

a. Type_____ b. Make_____

c. Speed range_____ d. Gear ratios; Forward 1_____

2_____ 3_____ 4_____ 5_____ 6_____ 7_____ 8_____

Reverse 1_____ 2_____ 3_____ 4_____

B. Transfer case.

a. Range_____ b. Type_____ c. Make_____

d. Speed ranges:_____ F _____ R e. Gear ratio: Forward

1_____ 2_____ 3_____ 4_____ 5_____ 6_____

Reverse 1_____ 2_____ 3_____ 4_____

V. Final drive.

a. No. driving wheels: Front_____ Rear_____

b. Type drive_____ c. Final drive gear ratio_____

d. Steering_____

VI. Front axle.

a. Lateral oscillation right side_____degrees,

left side_____degrees.

b. Steering type_____ Degrees articulation (if applicable)_____

c. Leaning wheel capability.

Left wheel

Lean to left _____ degrees.

Lean to right _____ degrees.

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Right wheel

Lean to left _____ degrees.

Lean to right _____ degrees.

d. Turning radius.

Right_____ Left_____

Turning radius is defined as the radius of the smallest circle described by the outside front wheel of the vehicle in making a 180-degree turn, measured from center to center of the outside front tire-thread ground imprint.

VII. Brakes.

1. Service brakes - primary system.

a. Type_____ b. Make_____ c. Model_____

d. Size_____.

e. Location_____ f. No. of wheels affected_____

g. Air compressor: Type_____ Size_____cfm Type Drive_____

Location_____ h. Reservoir size_____cu. in.

Pressure gage location_____ Operating pressure_____psi

2. Parking brake.

a. Type_____ b. Make_____ c. Model_____

d. Actuating_____wheels. e. Control: Type_____

located_____

VIII. Controls.

1. Operating.

a. Type: Open center_____ Closed center _____

Other_____

b. Overload protection_____

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2. Hydraulic system.

a. Filter: Make_____ b. Tank capacity _____ gal

Model_____

Element No. _____

c. Filler location _____ d. Oil Tank breather location_____

e. Pump: Type _____

Make_____

Model_____

Drive_____

Capacity_____ gpm at _____ rpm at _____ psi

f. Pressure relief valve:

Type_____ Make_____ Piston size_____

Pressure setting_____.

IX. Moldboard assembly.

1. Drawbar assembly.

a. Frame attachment_____.

2. Circle side shift.

a. Type_____ b. Distance: right_____in, left_____in.

3. Blade side shift.

a. Type_____ b. Distance: right_____in, left_____in.

4. End bits.

a. Type_____ b. Material_____.

5. Circle adjustments.

a. Type_____.

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6. Cutting edge.

a. Type_____ b. Material_____.

X. Operator's compartment and seat.

1. Compartment:

a. Material_____.

2. Seat:

a. Type_____ b. Dimensions_____.

3. Backrest:

a. Type_____ b. Dimensions_____.

XI. Scarifier or ripper.

a. Type_____ b. Mounting_____.

c. No. teeth_____ d. Tooth spacing_____in.

XII. Tires.

a. Type_____ b. Make_____ c. Size_____

d. Ply rating_____.

XIII. Electrical equipment.

1. Battery.

a. No._____ b. Size_____ c. Volts each_____

d. Make_____ e. Rating_____amp hrs, at 20-hr rate

f. No. plates_____ g. Mounting_____ h. Location_____

2. Alternator.

a. Type_____ b. Volts_____ c. Rating_____

watts at _____rpm d. Make_____ e. Model_____

f. Serial No._____ g. Drive_____ h. Control:

Type_____ Make_____ Model_____

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3. Horn.

a. Type_____ b. Make_____ c. Button location_____.

4. Starter.

a. Type_____ b. Make_____ c. Model_____

d. Serial No._____.

5. Electrical wiring.

a. Type_____ b. Protection_____.

XIV. Gages and instruments.

1. Ammeter or voltmeter.

a. Make_____ b. Range_____ c. Location_____

2. Odometer.

a. Make_____ b. Model_____ c. Location_____

3. Torque converter temp. gage.

a. Make_____ b. Model_____ c. Range_____

d. Location_____.

4. Torque converter pressure gage.

a. Make_____ b. Model_____ c. Range_____

d. Location_____.

5. Air pressure gage (if required).

a. Make_____ b. Range_____ c. Location_____

XV. Toolboxes.

a. Type_____ b. No._____ c. Dimensions_____

d. Location(s)_____ e. Cover fastening_____

XVI. Torque converter.

a. Make_____ b. Model_____ c. Serial No._____

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d. Type_____ e. Heat exchanger_____

f. Torque multiplication ratio_____ Oil pump_____

Oil filter_____.

XVII. Master clutch.

a. Make_____ b. Model_____ c. Type_____

d. Size_____ e. Facing size_____ f. Facing area_____

g. Torque capacity_____lb. ft.

h. Pilot bearing_____ i. Throwout bearing_____

j. Friction surface pressure_____psi.

XVIII. Fuel tank.

a. Capacity_____gal. b. Location_____

c. Drainplug location_____.

PART B. PHYSICAL AND FUNCTIONAL DIMENSIONS

XIX. Physical dimensions.

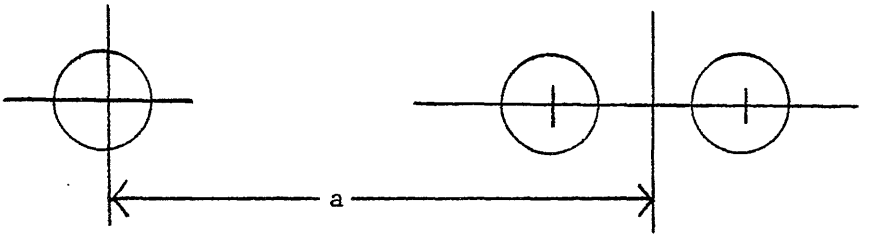
1. Operating weight with standard equipment. Front_____lb.

Rear_____lb. Total_____lb.

2. Operating weight with standard and optional equipment.

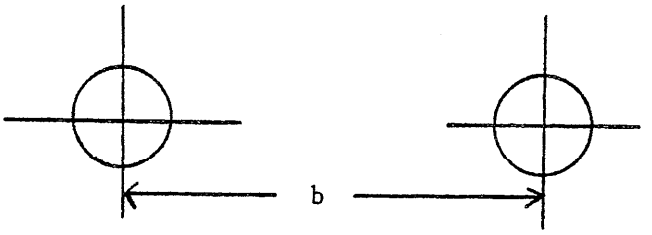
Front_____lb. Rear_____ lb. Total_____lb.

3. Wheelbase TYPES II, III, IV, & V



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TYPE I



a. _____ft. _____in. b. _____ft. _____in.

4. Scarifier.

a. Cutting width_____in. b. Weight_____lb.

c. No. teeth_____ d. Lift above ground_____in.

e. Scarifier down pressure _____lb.

5. Drawbar hitch.

a. Width_____in. b. Jaw opening_____in.

c. Pin diameter_____in.

XX. Functional dimensions.

1. Moldboard.

a. Cut below ground_____in. b. Lift above ground_____in.

c. Reach beyond tires: L_____ R_____ d. Lifting speed_____in./min. e. Lowering speed_____in./min.

f. Length with end bits_____ft. g. Width, measure on inside of curve_____ in. h. Thickness_____in.

i. Number of pitch positions _____.

XXI. Circle and moldboard positioning.

1. Rotation through _____ degrees (scarifier teeth removed).

2. Circle and moldboard positioning for bank sloping (see figure A and B).

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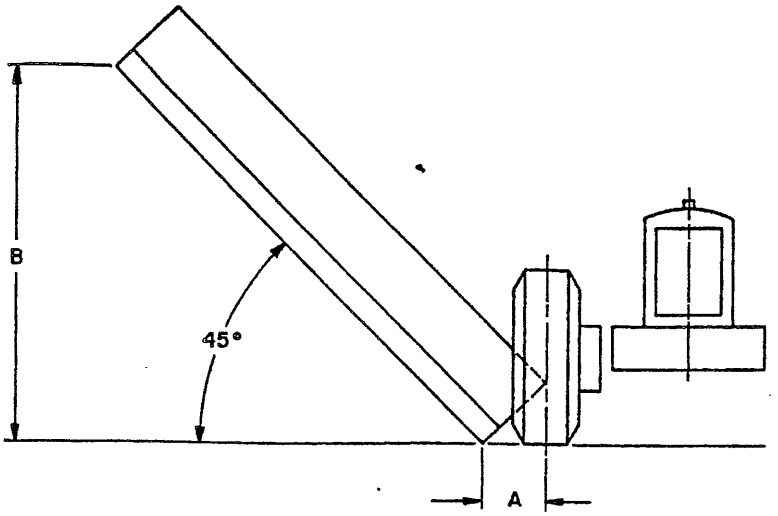


FIGURE A
SLOPING DIMENSIONS

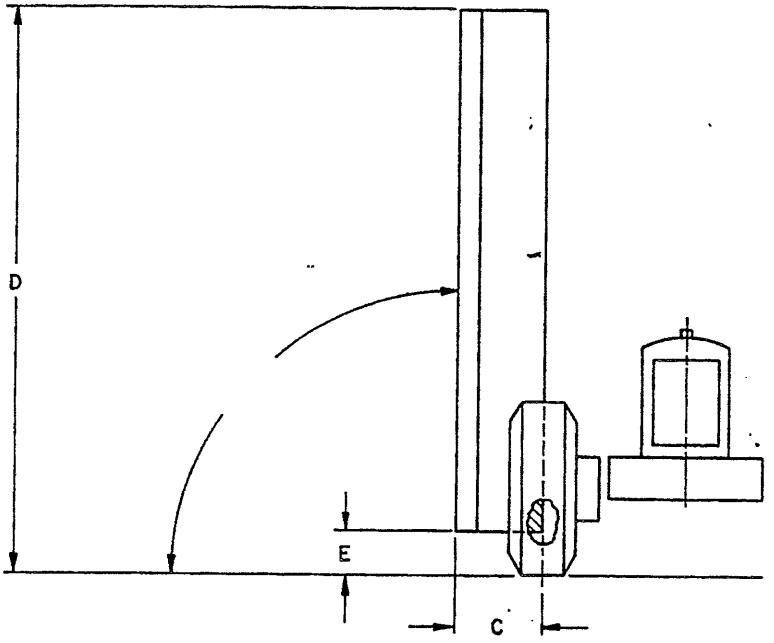


FIGURE B
EXTREME POSITION

CIRCLE AND MOLDBOARD
POSITIONING FOR BANK SLOPING

DX-1370

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Custodians

Army - ME
Navy - YD

Review activities

Army - AT
Navy - MC
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITIES:

DCG
NHT
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BLM
AFS

PREPARING ACTIVITY:

Army - ME

Project 3805-0119

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b. Recommended Wording:

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